KYLE DAWSON, PHD

PhD Applied Scientist | Mathematical Optimization, Machine Learning & Geospatial Modeling

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EXPERIENCE

Lead Data Scientist — Energy

Centerpoint Energy

Apr 2026 – Present Houston, TX

- lead a new geospatial intelligence (GEOINT) team, setting technical direction, mathematical-modeling and evaluation standards, and mentoring scientists and engineers across geospatial machine-learning and optimization initiatives
- designed and productionized a large-scale, spatially-constrained optimization model (mixed-integer linear programming via CBC with greedy heuristics, LightGBM risk scoring on LiDAR-derived features) that allocates a fixed operational budget across a geographic infrastructure network to maximize reliability (SAIFI) per dollar — an ambiguous business problem cast as a rigorous OR formulation on GCP
- built a deployed multi-LLM / generative-AI application (Gemini on Vertex AI, Cohere North) on Cloud Run with a retrieval benchmark scoring citation accuracy against a ground-truth set, to evaluate foundation models for document Q&A and image analysis
- ship production-grade Python systems (Docker, CI/CD, BigQuery) that carry spatial and ML models from research prototype through simulation and experimentation to reliable, large-scale production

Data Scientist → Senior Data Scientist — Climate Tech

SeekOps Inc.

Mar 2022 – Apr 2026 Austin, TX

- built and productionized machine-learning models to forecast survey quality, detect anomalies, and constrain uncertainty in gas-emission estimates, delivered through customer-facing analytics platforms (Plotly Dash on Azure) securing contracts of $50–150k
- implemented Bayesian uncertainty propagation and physics-based atmospheric boundary-layer and wind models to generate decision-grade metrics used in regulatory and enterprise reporting across 10+ countries
- invented novel measurement and quantification methodologies, presenting results to executives, customers, and international peer-reviewed conferences

Data Scientist — Ag Tech

Bayer CropScience

Jan 2020 – Mar 2022 St. Louis, MO

- applied machine learning, geospatial analytics, and weather- and soil-impact modeling to large-scale UAV- and aircraft-derived imagery for digital phenotyping and crop-performance assessment
- trained and evaluated deep-learning object-detection models (Detectron2) for classification from high-resolution georeferenced aerial imagery
- built end-to-end ML data workflows combining relational databases, geospatial data, and AWS S3 into machine-learning-ready datasets serving stakeholder decision support

Postdoctoral Fellow

NASA Langley Research Center

Feb 2018 – Jan 2020 Hampton, VA

- developed uncertainty-aware retrieval algorithms from remote-sensing observations and delivered peer-reviewed results, prototyping and experimenting with models against field data

SELECTED PROJECT

EXECUTIVE SUMMARY

PhD applied scientist (atmospheric science) with 8+ years building mathematical-optimization, machine-learning, and geospatial modeling systems for scientifically complex, large-scale operational problems. Pairs optimization (MILP/LP) and machine learning with large-scale geospatial data (LiDAR, satellite/aerial imagery, spatial infrastructure networks), weather and atmospheric modeling, and generative AI to turn ambiguous business problems into rigorous models and shipped software. Peer-reviewed researcher who carries solutions from prototype and simulation through experimentation into reliable cloud production, and communicates results to technical and business stakeholders.

CORE COMPETENCIES

Mathematical Optimization · Operations Research · Spatial Optimization · Research & Invention · Simulation & Experimentation · Technical Leadership · Cross-Functional Collaboration · Stakeholder Communication

DOMAIN EXPERTISE

Mathematical Optimization (MILP/LP) · Geospatial Analytics · LiDAR & Satellite Imagery · Machine Learning · Deep Learning · Atmospheric & Weather Modeling · Forecasting & Time Series · Generative AI & LLMs · Anomaly Detection · Bayesian Inference & Uncertainty · Computer Vision

TOOLS & TECHNOLOGIES

Python · PuLP / CBC · PyTorch · Scikit-Learn · LightGBM · XGBoost · Detectron2 · NumPy · Pandas · SQL · GCP BigQuery · Vertex AI · AWS · Azure · Docker · GPU / CUDA · MLOps / CI-CD · Git · GDAL · QGIS · Plotly · Anthropic Claude

BESS Day-Ahead Arbitrage Optimizer

Self-directed proof of concept

2025

A PuLP/CBC linear program, fed by ERCOT market API data, that schedules day-ahead battery charge/discharge arbitrage against nodal settlement-point (LMP) prices under state-of-charge and power constraints.

EDUCATION

PhD, Atmospheric Science

North Carolina State University

Graduated 2017 — Raleigh, NC

Contributions to the Understanding of Aerosol Microphysics Towards Improving the Assessment of Climate Radiative Forcing

AWARDS

NASA Postdoctoral Program Fellow

NASA Langley Research Center

Oct 2018 — Hampton, VA

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Langley Aerospace Research Scholar

NASA Langley Research Center

Jun 2013 — Hampton, VA

PUBLICATIONS

Google Scholar Profile